

MPC-MAP Project

Week 2 - Uncertainty

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1 Task 1 – Simulator

The simulator from the MPC-MAP project was successfully set up and explored. The main components of the system were analyzed:

- `setup.m` – defines the map and initial robot position
- `main.m` – runs the simulation loop
- `student_workspace.m` – main place for implementation
- `plan_motion.m` – defines robot control

Different maps and start positions were tested to understand the robot behavior.

2 Task 2 – Sensor Uncertainty

To analyze sensor uncertainty, the robot was placed in a static position (i.e., zero velocity) and sensor measurements were collected for more than 100 sampling periods.

The sampling period of the simulator is:

$$T_s = 0.1 \text{ s}$$

This corresponds to approximately 10 seconds of simulated time.

2.1 Implementation Details

During implementation, several important issues had to be addressed:

- LiDAR often returns `Inf` values when no obstacle is detected
- GNSS returns `NaN` values inside GNSS-denied areas
- Data had to be filtered to include only valid samples

To ensure correct data collection, the following rules were applied:

- LiDAR rows were accepted only if at least 6 out of 8 values were finite
- Infinite values were replaced with `NaN` and ignored during statistics
- GNSS samples were considered valid only if both X and Y values were finite

Additionally, a bug related to persistent variables was discovered. The GNSS data were unintentionally reset in every iteration because initialization was incorrectly tied to the LiDAR variable. This was fixed by initializing each persistent variable independently.

2.2 LiDAR Measurements

The LiDAR sensor provides distance measurements in 8 directions.

Indoor environments were used for LiDAR evaluation because obstacles are present in all directions, ensuring valid measurements.

Observed properties:

- Noise is approximately Gaussian
- Standard deviation is around 0.05
- Channels behave similarly and independently

2.2.1 Histogram

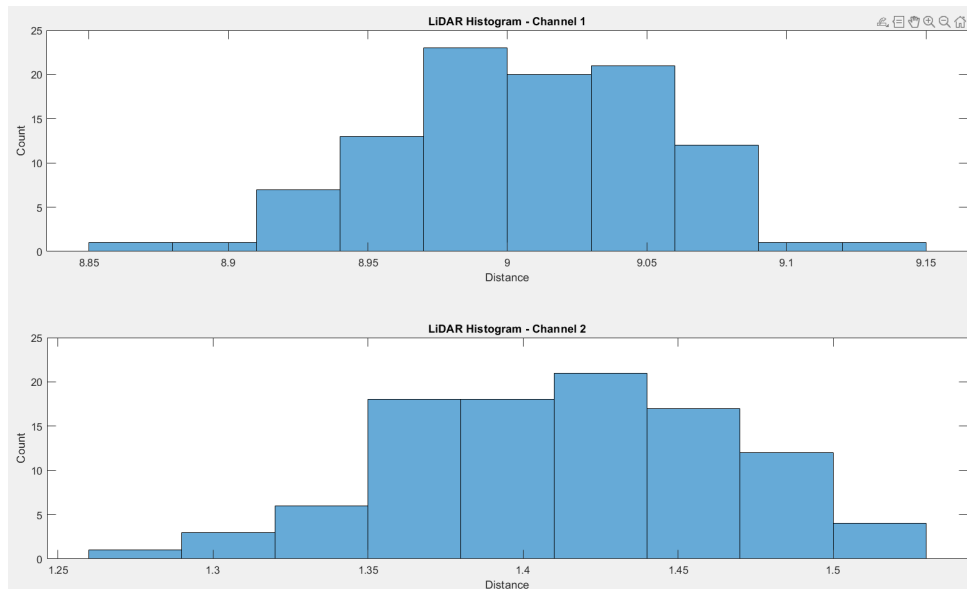


Figure 1: Histogram of LiDAR measurements

2.3 GNSS Measurements

GNSS provides noisy position measurements (x, y) .

Outdoor or mixed maps were used for GNSS evaluation to avoid GNSS-denied zones.

Observed properties:

- Measurements are centered around the true position
- Noise is approximately Gaussian
- Both axes are affected independently

2.3.1 Measured values

Example result:

Sigma GNSS:

0.4885 0.5242

2.3.2 Histogram

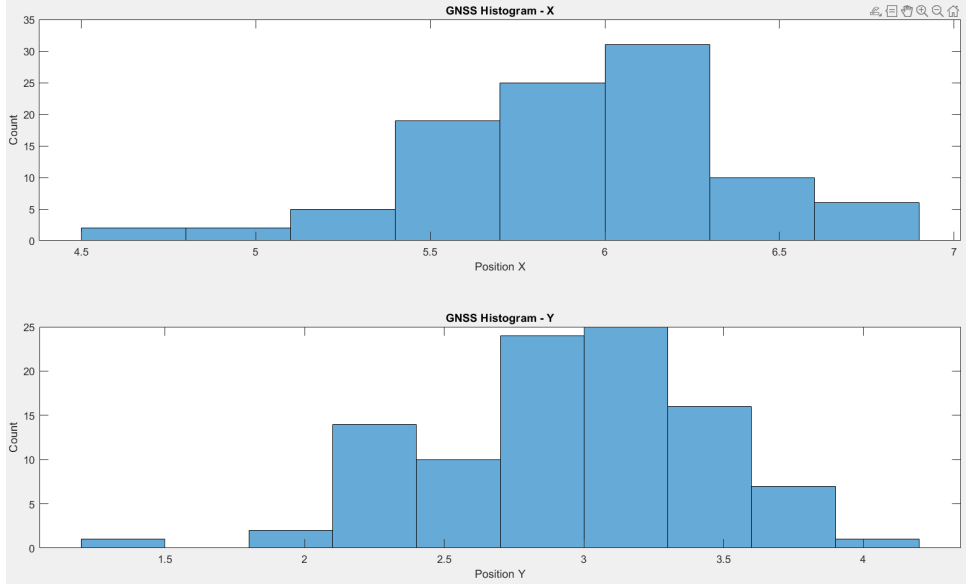


Figure 2: Histogram of GNSS measurements

3 Task 3 – Covariance Matrix

The covariance matrix describes the variability and correlation of measurements.

It was computed using MATLAB function:

```
cov(data)
```

3.1 Implementation Detail

Only valid samples were used for covariance computation:

- LiDAR: partial rows were used to ignore NaN values
- GNSS: only rows with both finite X and Y values were used

3.2 LiDAR Covariance

The covariance matrix has size 8×8 .

Observation:

- Diagonal elements correspond to variance (σ^2)
- Off-diagonal elements are close to zero
- Channels are nearly independent

3.3 GNSS Covariance

The covariance matrix has size 2×2 :

$$\begin{bmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{xy} & \sigma_y^2 \end{bmatrix}$$

Example result:

```
0.2386    -0.0114
-0.0114    0.2747
```

Observation:

- Variances match σ^2
- Correlation between axes is very small

4 Task 4 – Normal Distribution

A custom function was implemented to compute the probability density function:

```
function y = norm_pdf(x, mu, sigma)
    y = (1 ./ (sigma * sqrt(2*pi))) .* ...
        exp(-0.5 * ((x - mu) ./ sigma).^2);
end
```

4.1 Implementation Detail

Only one dimension was used for visualization (e.g., LiDAR channel 1 or GNSS X axis). The mean value was set to $\mu = 0$, and measured σ was used.

4.2 Results

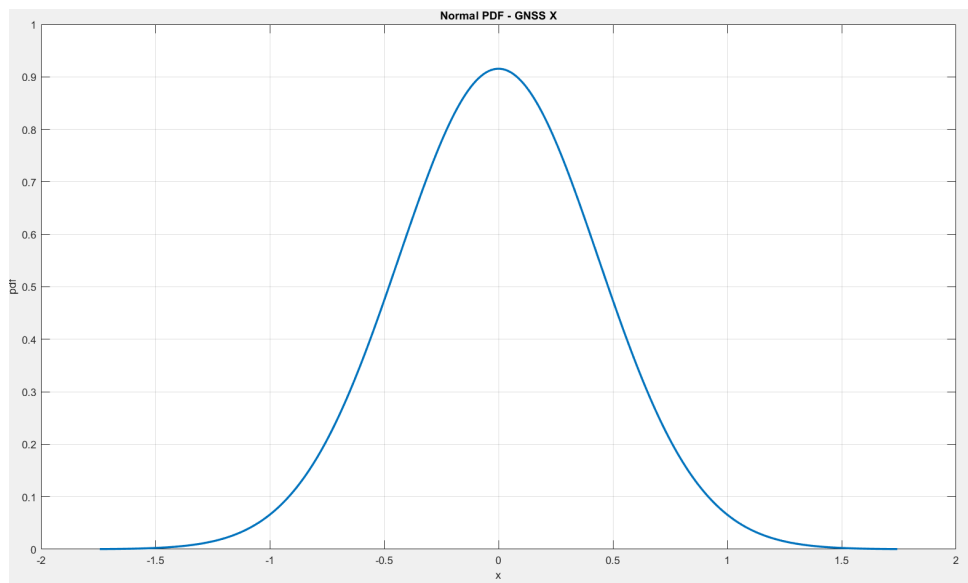


Figure 3: Normal distribution of sensor noise

Observation:

- The measured data closely follow the theoretical Gaussian curve
- This confirms that sensor noise can be modeled as a normal distribution

5 Task 5 – Motion Uncertainty

In this task, the robot was navigated to the goal without using any sensor feedback. Only predefined motion commands were used.

5.1 Control Strategy

The motion was defined as a sequence of wheel velocities depending on simulation time:

```
if k <= 50
    motion = [0.8, 0.8];
elseif k <= 80
    motion = [0.8, 0.8];
elseif k <= 90
    motion = [0.5, 0.8];
...
end
```

The commands were tuned manually based on the map layout.

5.2 Results

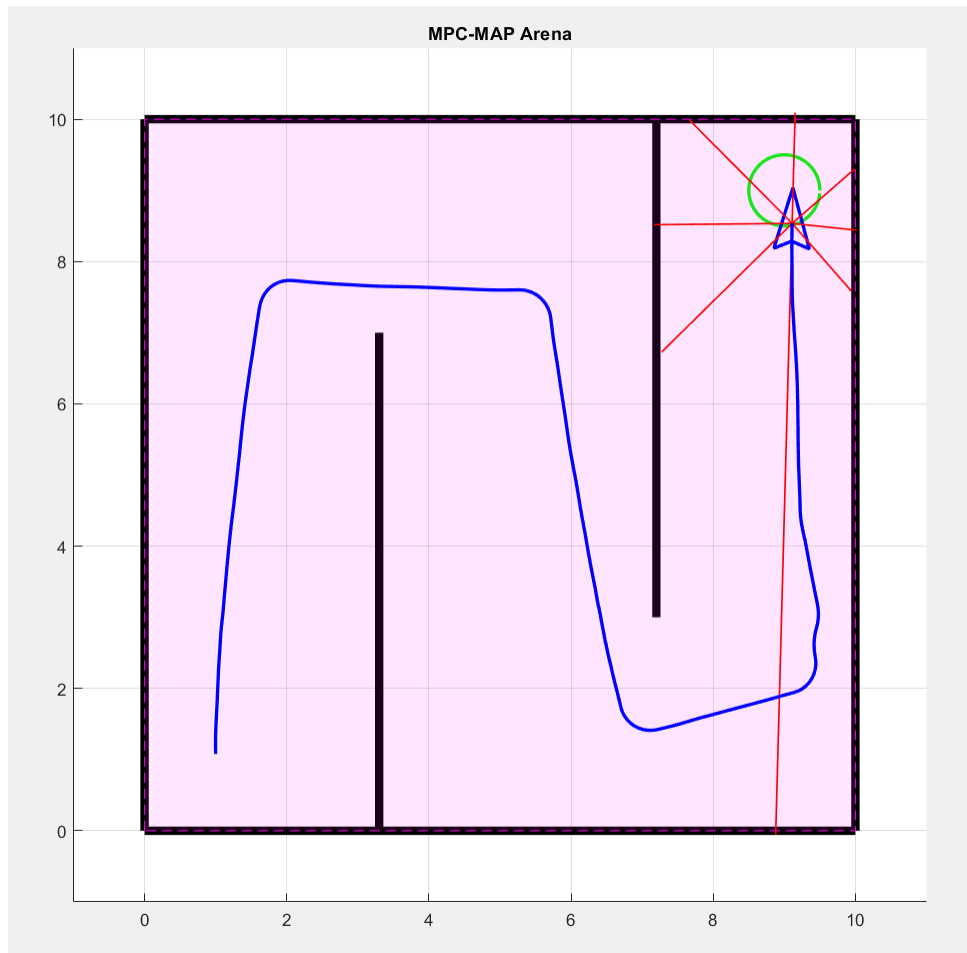


Figure 4: Robot trajectory without sensor feedback

Observation:

- The robot approximately reaches the goal
- The trajectory is not perfectly accurate
- Small errors accumulate over time

5.3 Sources of Uncertainty

Motion uncertainty arises from:

- Imperfect motion model
 - Numerical integration errors
 - Accumulation of small control inaccuracies
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6 Discussion

This experiment demonstrates that:

- LiDAR is reliable in environments with obstacles
 - GNSS is reliable in open environments
 - Each sensor has limitations
 - Sensor fusion is necessary for robust navigation
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7 Conclusion

The assignment showed that sensor measurements are inherently noisy and must be treated statistically.

Understanding uncertainty is essential for designing reliable robotic systems.

Combining multiple sensors allows overcoming individual limitations and improves overall system performance.